

ASSIGNMENT -1

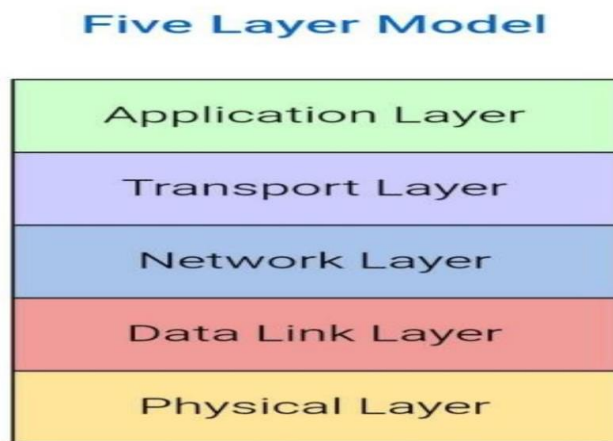
MODULE 1

COMPUTER NETWORKS

(PCCST501)

1. Draw the TCP/IP Protocol Stack and Explain the Role of Each Layer

The TCP/IP protocol suite is a five-layer architecture that enables communication between devices over the Internet. Each layer performs a specific function and provides services to the layer above it.



Application Layer

The application layer provides services directly to users and application programs. It enables communication between processes running on different hosts. Protocols such as HTTP, SMTP, FTP, DNS, and SSH operate at this layer.

Transport Layer

The transport layer provides process-to-process communication. It receives messages from the application layer, divides them into segments, and ensures reliable data delivery using TCP or fast communication using UDP.

Network Layer

The network layer is responsible for host-to-host communication. It uses the Internet Protocol (IP) to assign logical addresses and route packets through routers until they reach the destination.

Data Link Layer The data-link layer transfers frames between adjacent devices on the same network. It adds MAC addresses and prepares data for transmission across the physical medium.

Physical Layer

The physical layer transmits bits as electrical, optical, or wireless signals over cables or air.

2. Explain the Role of DNS in Converting the Domain Name into an IP Address

When the user types:

www.netflix.com

the browser cannot understand the domain name directly.

The Domain Name System (DNS) converts the domain name into an IP address.

Steps:

1. User enters **www.netflix.com**.
2. Browser checks local DNS cache.
3. If not found, the request goes to the DNS resolver.
4. DNS resolver contacts DNS servers.
5. The correct IP address is returned.
6. Browser connects to the Netflix server using that IP address.

Without DNS, users would need to remember numerical IP addresses instead of website names.

3. Describe How HTTP Is Used to Request and Receive the Webpage

Netflix mainly uses **HTTPS (HTTP Secure)**.

Process

1. Browser sends an HTTP GET request.

Example:

GET / HTTP/1.1

Host: www.netflix.com

2. Netflix server receives the request.
3. Server processes the request.
4. Server sends back:
 - HTML
 - CSS
 - JavaScript
 - Images
 - Videos (after login)
5. Browser renders the webpage.

HTTPS encrypts all communication to protect user accounts and payment information.

4.Explain How TCP Provides Reliable Communication for HTTP

- **Reliable delivery:** Ensures that Netflix webpages and streaming data reach the user's device correctly.
- **Error control:** Detects lost or damaged data and retransmits it to maintain smooth communication.
- **Flow control:** Prevents the Netflix server from sending data faster than the user's device can receive it.
- **Congestion control:** Adjusts the data transmission rate during network congestion to improve performance and reduce packet loss.

5.Identify Whether the Application Follows Client–Server or Peer-to-Peer Architecture and Justify Your Answer

Netflix follows a **Client–Server Architecture**.

Client

- User's browser or Netflix app

Server

- Netflix web servers and content delivery servers

Why Client–Server?

- Users request content.
- Netflix servers store and provide webpages and videos.
- Users do not share videos directly with one another.

Therefore, Netflix is a Client–Server application.

6.Compare HTTP and FTP

HTTP	FTP
Used for web browsing	Used for file transfer
Default port 80 (HTTPS uses 443)	Default ports 20 and 21
Transfers web pages	Transfers files between computers
Stateless protocol	Maintains a file transfer session
Supported by all web browsers	Requires an FTP client or browser support

7.Explain How Cookies Improve User Experience with One Real-World Example

Cookies are small text files stored in the user's browser.

They help websites remember user information.

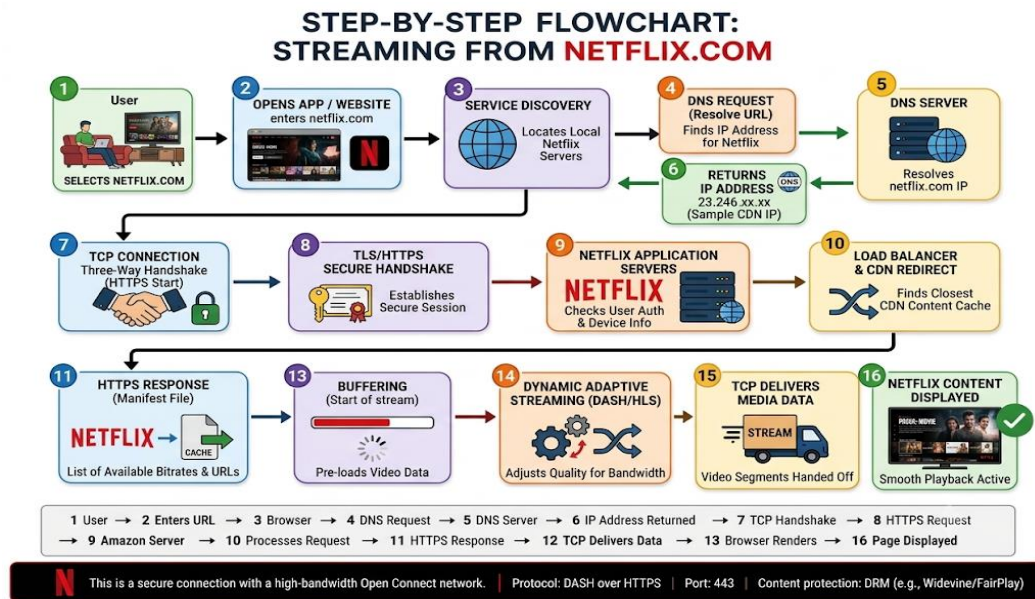
Netflix Example

When a user logs into Netflix:

- Login information is stored in cookies.
- Preferred language is remembered.
- Viewing history is saved.
- Personalized movie recommendations are shown.
- Users remain logged in during future visits.

Without cookies, users would need to log in every time they visited Netflix.

8. Flow Diagram Showing the Complete communication.



9. Write a brief conclusion (100–150 words) describing the importance of application-layer protocols in everyday Internet usage.

Application-layer protocols are essential for enabling smooth and secure communication on the Internet. Whenever a user visits Netflix, several protocols work together to deliver the webpage efficiently. DNS converts the website name into an IP address, TCP establishes a reliable connection and ensures accurate data delivery, and HTTPS securely transfers webpage content between the browser and the server. Cookies further enhance the browsing experience by remembering login details, user preferences, and personalized recommendations. These technologies operate seamlessly in the background, allowing users to access websites quickly, safely, and conveniently. Understanding how these protocols interact provides valuable insight into the functioning of modern computer networks and highlights their importance in everyday Internet activities such as streaming, online shopping, communication, and web browsing.